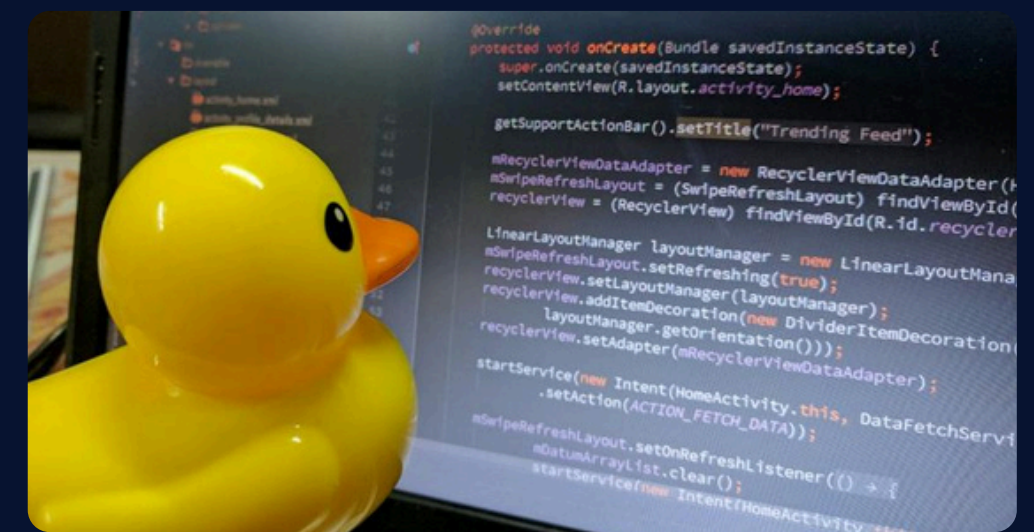




Important Statistical Concepts



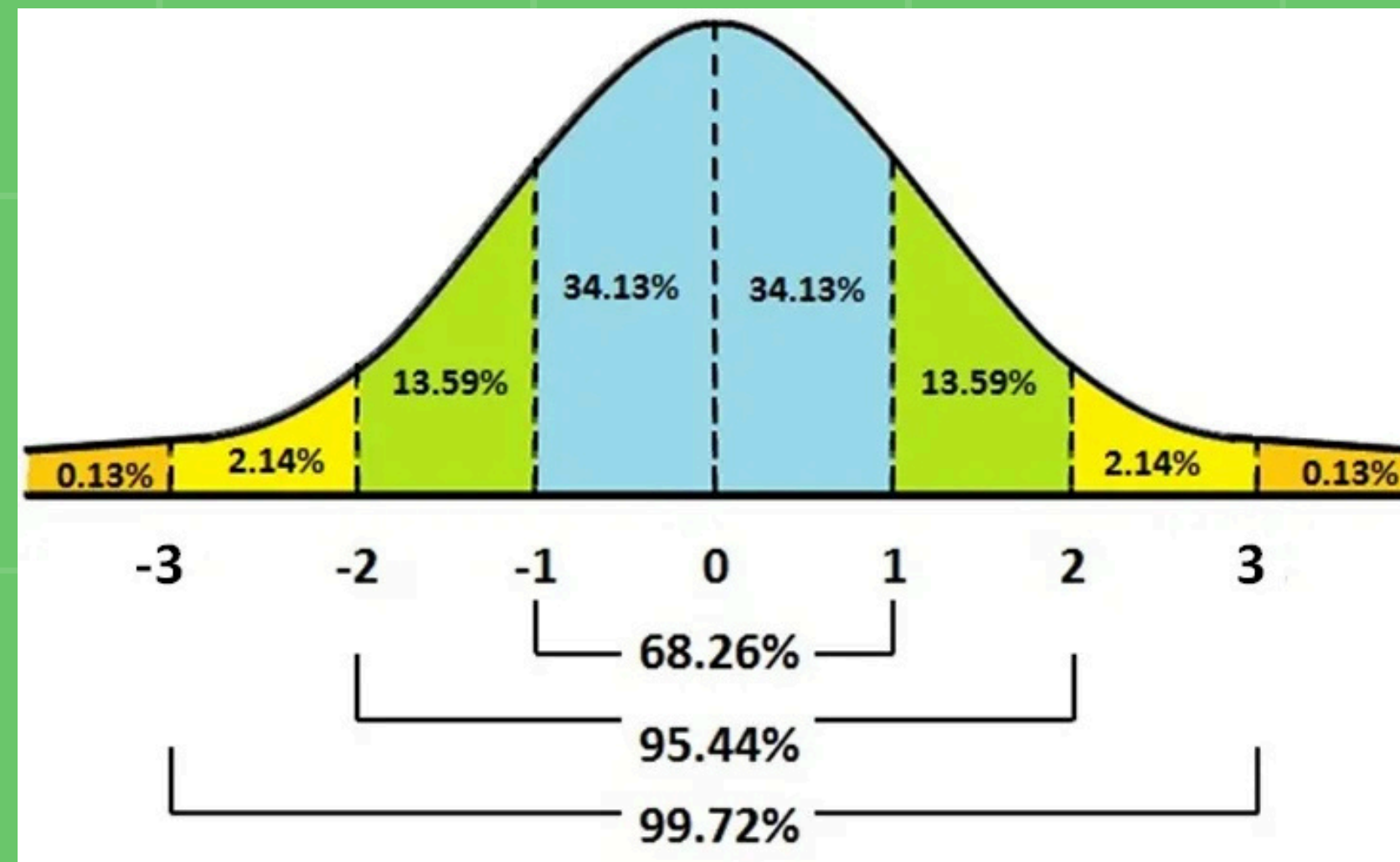
Session -> 6



Normal Distribution

What is Normal Distribution?

- The normal distribution (or Gaussian distribution) is a bell-shaped curve.
- In machine learning and statistics, the normal distribution plays a central role – many algorithms assume or work better when data is normally distributed



Properties of Normal Distribution

- Mean = Median = Mode (all at the center).
- Data is symmetrical around the mean.
- 68% of data falls within ± 1 standard deviation (σ).
- 95% within $\pm 2\sigma$, and 99.7% within $\pm 3\sigma$ (Empirical Rule).

Example:

If we have students' heights in a normal distribution:

- mean = 170
- standard deviation = 10

>>Then 68% of students their heights between 160 and 180

Detecting Outliers

In the previous example, if we follow the rule of normal distribution:

- 68% of students $\approx 170 \pm 10$ cm (160–180 cm)
- 95% $\approx 170 \pm 20$ cm (150–190 cm)
- 99.7% $\approx 170 \pm 30$ cm (140–200 cm)

Then the outliers are greater than 200 cm and shorter than 140 cm.



Descriptive Statistics

Mean, Median, and Mode (central tendency measures)

- they give us a single value that represents our dataset

1- Mean:

mean = sum of observation / total number of observations

Example:

data = 10, 20, 10, 40, 30

mean = $(10+20+10+30+30) / 5 = 20$

Mean, Median, and Mode

(central tendency measures)

Problem of the mean:

The mean is sensitive to outliers, so it can be misleading and do a poor representative for our data.

example: most of the salaries are between 5000 and 6000, but the CEO's salary is 25000, so the mean will be away from the mid of the data.

- ✓ Use mean when data is balanced.
- ✗ Avoid mean if outliers exist.

Mean, Median, and Mode (central tendency measures)

2- Median(Middle Value):

median = The middle number in an ordered dataset.

Example:

data = 10, 20, 10, 40, 30

median: first order the data → 10, 10, 20, 30, 40

the median is the middle value, which is 20

Mean, Median, and Mode (central tendency measures)

2- Median(Middle Value):

Note: if the number of data is even not odd, then there is no middle value → Take the two values in the middle, sum them together, then divide by 2.

Example: 1, 2, 3, 4

$$\text{median} = (2+3)/2 = 2.5$$

✓ Use median when outliers may exist (e.g., income, house prices).

Mean, Median, and Mode

(central tendency measures)

3- Mode (Most Frequent Value):

Mode = the value that has the max count in the data
(appears most often)

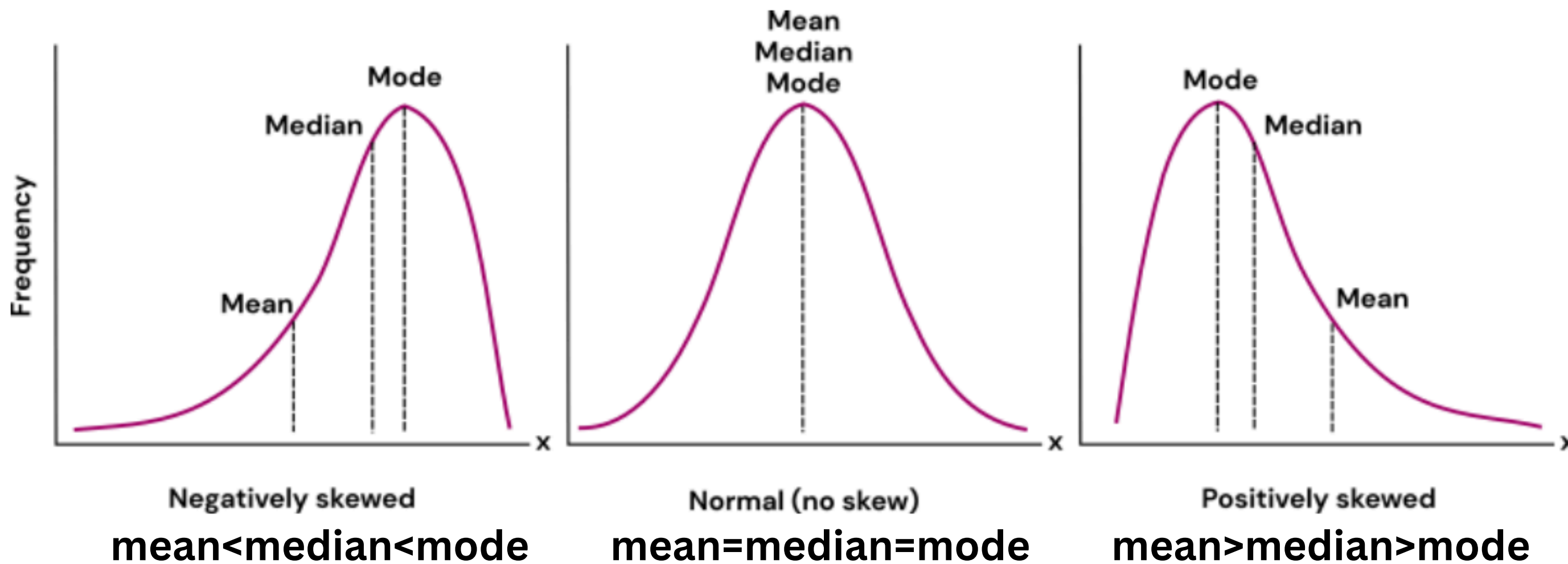
- It is used for categorical data

Example:

```
data = Vanilla, Chocolate, Chocolate, Mango, Chocolate  
mode = Chocolate
```

- ✓ Use mode for categories or repeating values.

Mean, Median, and Mode (central tendency measures)



Variance

- is a measure of how far a set of data are dispersed out from their mean or average value.

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

- The variance is 100 = On average, the squared difference between each data point and the mean is 100.
- **Q**: What does it mean when I say variance is 0?

Standard Deviation

- The standard deviation (SD) is the square root of the variance:

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

- A **larger standard deviation** means more spread – the data points are more scattered from the mean.

Variance **vs** Standard deviation

✓ We often use standard deviation instead of variance because:

It removes the square from the units, making it easier to interpret.

Example:

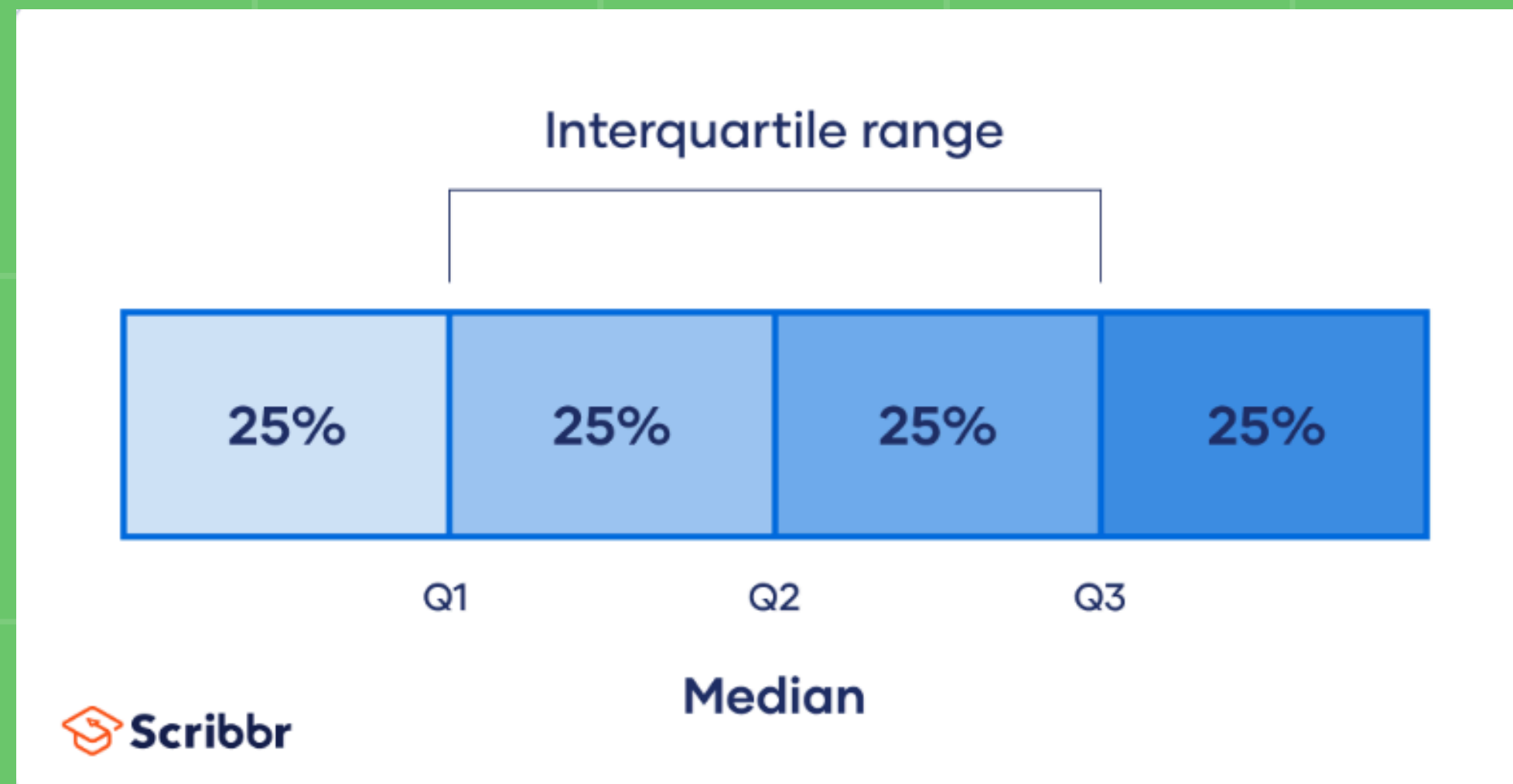
- You can't say "this person spends 20,000 dollars squared more" – it's not a real-world unit.-
- instead, we use standard deviation to day: On average, people spend about \$141.42 more or less than the mean."



Interquartile Range (IQR)

Interquartile Range (IQR)

- It is a measure of statistical dispersion
- We use it to figure out the distance between the middle of the first half and the middle of the second half.



Interquartile Range (IQR)

Example: Find the IQR for the following dataset:

4, 4, 10, 11, 15, 7, 14, 12, 6

1- sort the data:

4, 4, 6, 7, 10, 11, 12, 14, 15

2- find the median:

4, 4, 6, 7, **10**, 11, 12, 14, 15

3- Find the median of the 1st half (Q1):

$$(4 + 6)/2 = 5$$

4- Find the median of the 2nd half (Q3):

$$(12 + 14)/2 = 13$$

Interquartile Range (IQR)

Example: Find the IQR for the following dataset:

4, 4, 10, 11, 15, 7, 14, 12, 6

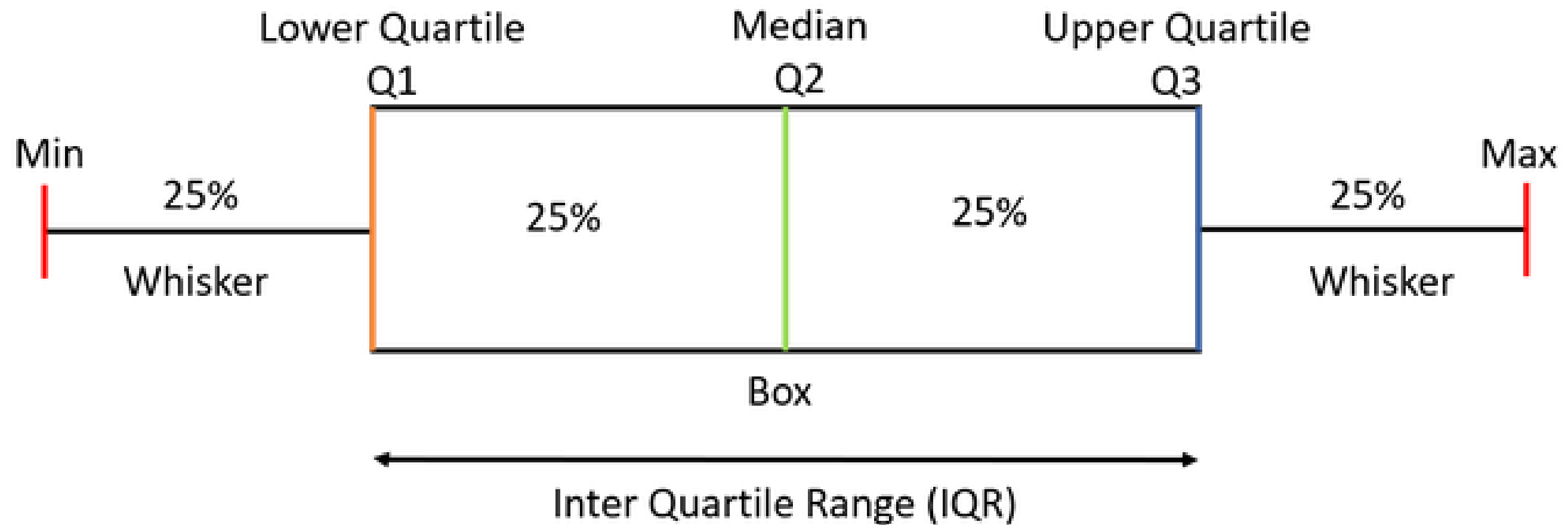
5- Calculate IQR → difference between the middle of the first half and the middle of the second half:

$$13 - 5 = 8$$

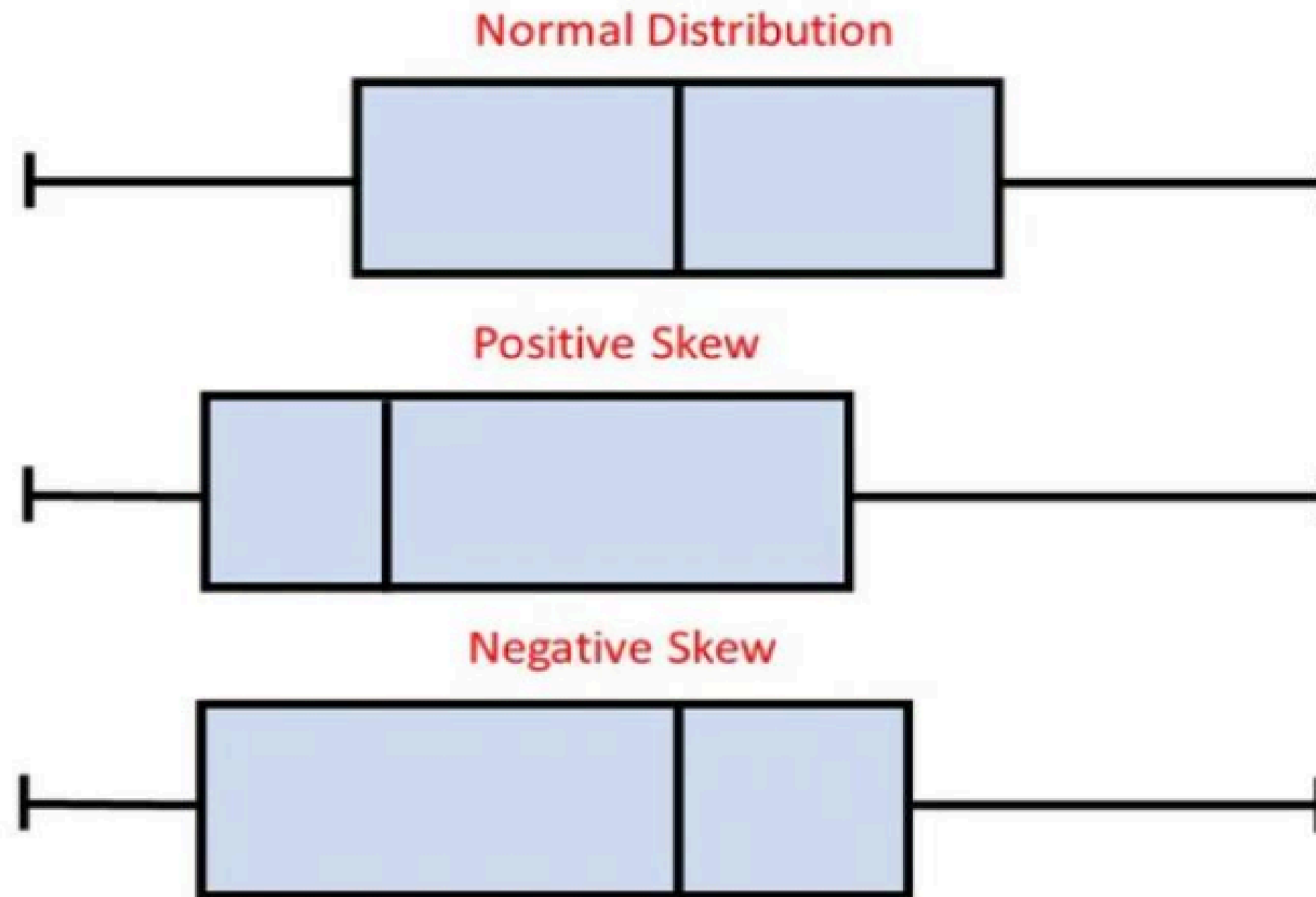
How Can IQR identify outliers?

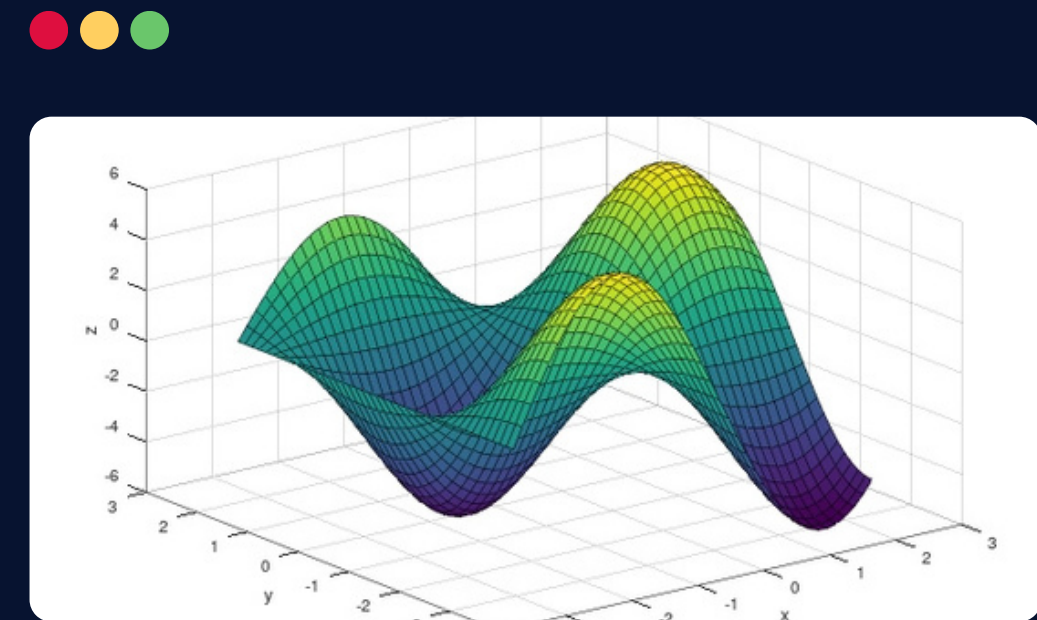
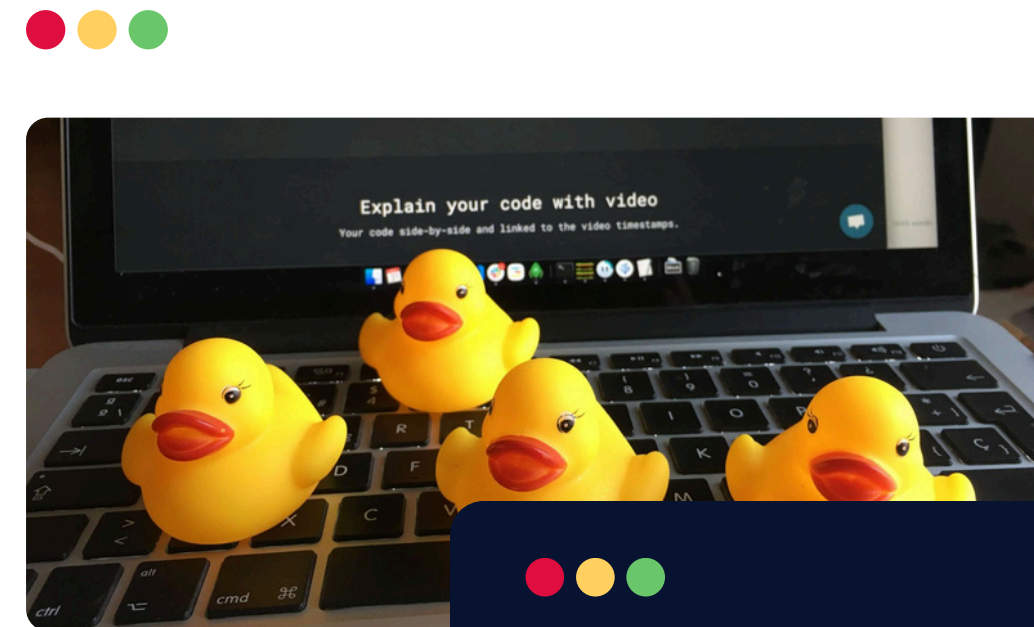
- We create boundaries called “Fences”:
 - Lower Fence:
 $Q1 - 1.5 * IQR$
 - Upper Fence:
 $Q3 + 1.5 * IQR$
- Any points that fall below the lower fence or above the upper fence are considered outliers.

Box Plot



Box Plot





Thank You

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